Two-Way ANOVA

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Introduction

Two-way ANOVA tests the effects of two categorical factors on a continuous outcome, plus their interaction. It generalises one-way ANOVA by accommodating designs that manipulate or measure two conditions simultaneously, and is the simplest example of a factorial design.

Prerequisites

One-way ANOVA, factorial designs.

Theory

For a two-way design with factors A and B:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk},$$

where $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. The ANOVA partitions the total sum of squares into:

- Main effect of A (df = a - 1).
- Main effect of B (df = b - 1).
- Interaction AxB (df = (a - 1)(b - 1)).
- Residual (df = ab(n - 1) for balanced design with n per cell).

Three null hypotheses are tested: no main A effect, no main B effect, no interaction. The F-tests use the residual MS as denominator.

Type I, II, III sums of squares differ when the design is unbalanced:

- Type I: sequential; order-dependent.
- Type II: each effect adjusted for others except interactions involving it.
- Type III: each effect adjusted for all others; default in most modern practice when interactions are present.

Assumptions

- Independent observations.
- Normal residuals.
- Homogeneous variances across cells.
- For balanced designs all SS types agree; for unbalanced ones, the choice matters.

R Implementation

```r
library(car); library(effectsize); library(emmeans)
set.seed(2026)

df <- expand.grid(A = factor(c("A1", "A2")),
                  B = factor(c("B1", "B2", "B3")),
```

```r
                  rep = 1:20)
df$y <- with(df, 50 + 3*(A == "A2") + 2*(B == "B2") + 4*(B == "B3") +
                5*(A == "A2" & B == "B3") + rnorm(nrow(df), 0, 6))

# Fit and Type III ANOVA
fit <- lm(y ~ A * B, data = df, contrasts = list(A = contr.sum, B = contr.sum))
Anova(fit, type = 3)

# Effect sizes
eta_squared(fit, partial = TRUE)

# Marginal means and interaction plot
emm <- emmeans(fit, ~ A * B)
emm
pairs(emm, simple = "A")

# Simple interaction plot
interaction.plot(df$B, df$A, df$y, col = c("#2A9D8F", "#F4A261"),
                 lwd = 2, main = "Interaction plot")
```

Output & Results

```
Anova Table (Type III tests)

            Sum Sq  Df  F value  Pr(>F)
(Intercept) 291 000  1  8036    <2e-16
A             168   1   4.64  0.0332
B             490   2   6.77  0.0017
A:B           186   2   2.57  0.0819
Residuals   4 123 114

Eta2 (partial) | 95% CI
-------------------------
A              | 0.04
B              | 0.11
A:B            | 0.04
```

Main effects of A and B are significant; the interaction is borderline.

Interpretation

"In a two-way ANOVA, both A ($F(1, 114) = 4.64$, $p = 0.033$) and B ($F(2, 114) = 6.77$, $p = 0.002$) had significant main effects; the A x B interaction was not significant ($F(2, 114) = 2.57$, $p = 0.082$)."

Practical Tips

- Use Type III SS with sum-to-zero contrasts for standard interpretation.
- Always inspect the interaction plot; visual patterns often matter more than p-values.
- Unbalanced designs need careful contrast specification; `car::Anova` handles the SS types explicitly.
- Report effect sizes (partial η^2) alongside F.
- For significant interactions, interpret simple effects, not the main effects alone.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests